Random walks with square-root boundaries.

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Let $\{X_n\}$ be a sequence of independent, identically distributed random variables. We shall always assume that the random variables have zero mean and finite variance.

Let $S(n)$ be a random walk with increments X_k, that is,

$$S(n) := X_1 + X_2 + \ldots + X_n, \quad n \geq 1.$$

For every function $g(t)$ we define by T_g the first time when S down-crosses g, i.e.

$$T_g := \inf\{n \geq 1 : S(n) \leq g(n)\}.$$

We want to understand the asymptotic behaviour of $\mathbf{P}(T_g > n)$ in the case when $g(t) \sim c\sqrt{t}$ as $t \to \infty$.

The most classical boundary crossing problem for random walks deals with constant boundaries $g(t) \equiv -x$, $x \geq 0$. In this case one has

$$\mathbf{P}(T_x > n) \sim V(x) n^{-1/2},$$

where $V(x)$ is harmonic for S killed at time T_x:

$$V(x) = \mathbf{E}[V(x + X_1); x + X_1 > 0].$$

The standard approach is based in the Wiener-Hopf factorisation, which works properly only for constant boundaries.

Mogulskii and Pecherskii (1978) have shown that if the boundary satisfies the condition $g(n+k) \leq g(n) + g(k)$ then there exists the sequence of events $\{E_n\}$ such that

$$\sum_{n=0}^{\infty} z^n \mathbf{P}(T_g > n) = \exp\left\{\sum_{n=1}^{\infty} \frac{z^n}{n} \mathbf{P}(E_n)\right\}.$$

But the events E_n are not easy to analyse, and it is not clear how to determine the asymptotic behaviour of $\mathbf{P}(E_n)$.

Denisov, Sakhanenko and W. (2019): If $g(t) = o(\sqrt{t})$ then there exists a positive, slowly varying function L_g such that

$$\mathbf{P}(T_g > n) \sim \frac{L_g(n)}{n^{1/2}}.$$

One can describe L_g in terms of the harmonic function V:

$$L_g(n) = \mathbf{E}[V(S(n) - g(n)); T_g > n].$$

Moreover, if $g(t)$ is monotone then $\lim_{n\to\infty} L_g(n) \in (0, \infty)$ if and only if $\int_1^\infty \frac{|g(t)|}{t^{3/2}} < \infty$.

The proof is based on the following universality idea: after Brownian scaling there is almost no differences between $g(t)$ and constant boundaries.

If $g(t) \sim c\sqrt{t}$ with some $c \neq 0$ then one can not expect that the boundary should be similar to the constant one.

But we have a more standard universality: every random walk with zero mean and finite variance converges, after rescaling, towards the Brownian motion. Thus, the behaviour of the exit time for a random walk should be similar to that of the corresponding exit time for the Brownian motion.

Let $B(t)$ be a standard Brownian motion and let

$$T_g^{bm} := \inf\{t > 0 : B(t) \le g(t)\}.$$

For every $b \ge 0$ and $a > c\sqrt{b}$ we define

$$g_{a,b}(t) := c\sqrt{t+b} - a$$

and set, for brevity,

$$T_{a,b}^{bm} := T_{g_{a,b}}^{bm}$$

Let $D_p(z)$ denote the parabolic cylinder function. We define

$$\psi_p(x) := e^{x^2/4} D_p(x) \quad \text{and} \quad V_p(x,t) := t^{p/2} \psi_p\left(\frac{x}{\sqrt{t}}\right), \quad t > 0.$$

These functions satisfy

$$\left(\frac{\partial}{\partial t} + \frac{1}{2}\frac{\partial^2}{\partial x^2}\right) V_p(x,t) = 0.$$

Novikov (1971) has shown that

$$\text{(1)} \qquad \mathbf{E}(T_{a,b}^{bm} + b)^{p/2} = \begin{cases} \frac{V_p(a,b)}{\psi_p(c)}, & p < p(c), \\ \infty, & p \geq p(c), \end{cases} \qquad b \geq 0,$$

where $p(c)$ is the minimal positive root of the mapping $p \mapsto \psi_p(c)$, i.e. $\psi_{p(c)}(c) = 0$.

Analysing this Mellin transform at the critical point $p(c)$ one can show that

$$\mathbf{P}(T_{a,b}^{bm} > t) \sim \kappa(c) \frac{V_{p(c)}(a,b)}{t^{p(c)/2}} \quad \text{as} \quad t \to \infty$$

for some positive constant $\kappa(c)$.

Uchiyama (1980) has shown that

$$\lim_{t\to\infty} t^{p(c)}\mathbf{P}(T_g^{bm} > t) \in (0,\infty)$$

provided that $|g(t) - c\sqrt{t}|$ remains bounded.

Denisov, Hinrichs, Sakhanenko and W. (2022):

$$\mathbf{P}(T_g^{bm} > t) \sim \kappa(c)\frac{U_g(t)}{t^{p(c)/2}}$$

with

$$U_g(t) = \mathbf{E}[V_{p(c)}(B(t) - g(t) + c\sqrt{t}, t); T_g > t].$$

U_g is slowly varying.

The proof of the latter result uses the following observation made by Novikov: the process $V_{p(c)}(a + B_t, b + t)1\{T_{a,b}^{bm} > t\}$ is a martingale.

In other words, $V_{p(c)}$ is a positive space-time harmonic function for $B(t)$ killed at crossing $g_{a,b}$.

Can one construct such a function for discrete time walk $S(n)$?

Theorem 1. Assume that $\mathbf{E}|X_1|^{p(c)} < \infty$ and $\mathbf{E}|X|^{2+\delta} < \infty$ for some $\delta > 0$. Then the function

$$W(a,b) := \lim_{n\to\infty} \mathbf{E}[V_{p(c)}(a+S(n), b+n); T_{a,b} > n]$$

is well-defined and satisfies

$$W(a,b) = \mathbf{E}[W(a+S(n), b+n); T_{a,b} > n], \quad n \geq 1.$$

Corollary 2. Under the conditions of Theorem 1,

$$\mathbf{P}(T_{a,b} > n) \le M \frac{W(a,b)}{n^{p(c)/2}}, \quad n \ge b,$$

with some constant M which is independent of a and b.

To prove Theorem 1 we first construct a family of supermartingales for the killed random walk:

For every $p < p(c)$ set

$$f(x,t) := \left(V_p(x+R,t) - C \cdot |x+R|^{p-\delta}\right)_+ .$$

Then there exist constants C and R such that the sequence

$$f(a + S(T_{a,b} \wedge n), b + T_{a,b} \wedge n)$$

is a supermartingale.

$V_p(a + B(T^{bm}_{a,b} \wedge t), T^{bm}_{a,b} \wedge t)$ is a martingale. For discrete time walks we need a corretion, which is of smaller order than V_p.

These supermartingales allow one to obtain the following intermediate bounds:

For every $p < p(c)$ there exists a finite constant C such that for all a, b,

$$\mathbf{P}(T_{a,b} > n) \le C \frac{1 + |a|^p + b^{p/2} 1\{p(c) < 1\}}{n^{p/2}}.$$

Moreover, for every $p < p(c)$,

$$\mathbf{E}\left[\left(\max_{k \le T_{a,b}} |S(k)|\right)^p\right] \le C(1 + |a|^p + b^{p/2} 1\{p(c) < 1\}).$$

Having these bounds, one can adjust the construction of harmonic functions from Denisov and W. (2015) to the case of square root boundaries. As a result one has a space-time harmonic function $W(a,b)$.

The main idea is again to use universality: harmonic function for $B(t)$ is the right 'initial point' for iterations.

Theorem 3. Assume that $\mathbf{E}[X] = 0$, $\mathbf{E}[X^2] = 1$ and that $\mathbf{E}[|X|^{2+\delta}]$ is finite for some $\delta > 0$. We also assume that $\mathbf{E}[|X|^{p(c)}] < \infty$ and in the case when $\limsup_{t\to\infty}(g(t) - c\sqrt{t}) = \infty$ we assume that $\mathbf{E}[|X|^{p(c)+\delta}] < \infty$. Then

$$\mathbf{P}(T_g > n) \sim \kappa \frac{U_g(n)}{n^{p(c)/2}},$$

where the function $U_g(n) := \mathbf{E}[W(S(n), n); T_g > n]$ is slowly varying. In the special case $g(t) = c\sqrt{t+b} - a$ with $a > c\sqrt{b}$ one has

$$\mathbf{P}(T_g > n) \sim \kappa(c) \frac{W(a, b)}{n^{p(c)/2}}.$$

In the case $p(c) > 2$ our moment assumption $\mathbf{E}|X|^{p(c)}$ seems to be almost optimal:

Assume that $\mathbf{P}(X > u) \sim c_0 u^{-p}$ with some $p < p(c)$. Then, by the central limit theorem,

$$\mathbf{P}(T_g > n) \geq c_1 \mathbf{P}(X_1 > (c+1)\sqrt{n}) \sim c_2 n^{-p/2}.$$

Therefore, the tail of T_g may decay slower if $\mathbf{E}|X|^{p(c)-\delta} = \infty$ for some $\delta > 0$.

Greenwood and Perkins (1983) have shown that if $c < 0$ ($p(c) < 1$) then

$$\mathbf{P}(T_g > n) \sim \frac{L_g(n)}{n^{p(c)/2}}$$

provided that $\mathbf{E}X^2 \log|X| < \infty$.

Their moment condition is a bit weaker, since we had to assume that $\mathbf{E}|X|^{2+\delta}$ is finite.

Theorem 4. Assume that $\mathbf{E}[X]=0$, $\mathbf{E}[X^2]=1$ and that $\mathbf{E}[|X|^2\log^{2+\delta}X]$ is finite for some $\delta>0$. We also assume that $\mathbf{E}[|X|^{p(c)}]<\infty$. Then the following statements hold.

(a) The space-time harmonic function from Theorem 1 is well-defined.

(b) In the case $g(t)=c\sqrt{t+b}-a$ with $a>c\sqrt{b}$ one has

$$\mathbf{P}(T_{a,b}>n)\sim\kappa(c)\frac{W(a,b)}{n^{p(c)/2}}.$$

(c) If $\int_1^\infty t^{-3/2}\max_{s\le t}|g(s)-c\sqrt{s}|<\infty$ then

$$n^{p(c)/2}\mathbf{P}(T_g>n)\to C_g\in(0,\infty).$$

This time the form of the supermartingale is much more complicated. We consider

$$V_{p(c)}(x + 2h(n+1), n+1) + Rv(x + 2h(n+1), n+1),$$

where $v(x,t)$ is the solution to

$$\begin{cases} \left(\frac{\partial}{\partial t} + \frac{1}{2}\frac{\partial^2}{\partial x^2}\right) v(x,t+1) = -h(t+1)V_{p(c)-1}(x,t+1)),\ x > c\sqrt{t+1}, \\ v(c\sqrt{t+1}, t+1) = 0 \end{cases}$$

$h(t)$ is a regularly varying smooth majorant for $g(t) - c\sqrt{t}$.